

$$(a) \quad I_{D6} = I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV6})^2$$

$$160 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times 40 \times (V_{OV6})^2$$

$$\therefore V_{OV6} = 0.2 \text{ V}$$

$$V_{in,CM,min} = V_{OV5} + V_{GS1}$$

$$0.7 = 0.2 + V_{GS1}$$

$$\therefore V_{GS1} = 0.5 \text{ V}$$

$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV1})^2$$

$$80 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times \frac{W}{L} (0.5 - 0.4)^2$$

$$\left(\frac{W}{L}\right)_{1,2} = 80$$

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{thn}$$

$$1.55 = 1.8 - |V_{GS3}| + 0.4$$

$$|V_{GS3}| = 0.65 \text{ V}$$

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV3})^2$$

$$80 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times \frac{W}{L} \times (0.15)^2$$

$$\left(\frac{W}{L}\right)_{3,4} = 71.11$$

$$(b) \quad A_v = g_{m1} (r_{o1} \parallel r_{o3})$$

$$g_{m1} = \frac{2I_{D1}}{V_{OV1}} = \frac{2 \times 80 \times 10^{-6}}{0.1} = 1.6 \text{ mS}$$

$$r_{o1} = \frac{1}{\lambda_n I_D} = \frac{1}{0.05 \times 80 \times 10^{-6}} = 250 \text{ k}\Omega$$

$$r_{o3} = \frac{1}{\lambda_p I_D} = \frac{1}{0.1 \times 80 \times 10^{-6}} = 125 \text{ k}\Omega$$

$$\therefore A_v = 1.6 \text{ mS} (250 \text{ k} \parallel 125 \text{ k})$$

$$= 1.6 \text{ mS} \times 83.33 \text{ k} = 133.33 \text{ V/V}$$

$$(c) \quad \text{Output voltage swing} = V_{DD} - |V_{OV4}| - V_{OV2} - V_{OV5}$$

$$= 1.8 - 0.15 - 0.1 - 0.2$$

$$= \underline{\underline{1.35 \text{ V}}}$$

$$(d) \quad f_{3-dB} = \frac{1}{2\pi C_L R_{out}}$$

$$= \frac{1}{2\pi \times 3 \times 10^{-12} \times 83.33 \text{ k}} = 636.64 \text{ kHz}$$

$$(e) \quad f_{3-dB, closed} = f_{3-dB, open} (1 + \beta A)$$

$$= 636.64 (1 + 133.33) \text{ kHz} = 85.52 \text{ MHz}$$

or

$$f_{3-dB, closed} = \frac{1}{2\pi C_L \left(\frac{1}{g_{m2}} \parallel r_{o4}\right)} = 85.31 \text{ MHz}$$

$$(a) \quad I_{D6} = I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV6})^2$$

$$200 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times 32 \times (V_{OV6})^2$$

$$\therefore V_{OV6} = 0.25 \text{ V}$$

$$V_{in,CM,min} = V_{OV5} + V_{GS1}$$

$$0.8 = 0.25 + V_{GS1}$$

$$\therefore V_{GS1} = 0.55 \text{ V}$$

$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV1})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times \frac{W}{L} (0.55 - 0.4)^2$$

$$\left(\frac{W}{L}\right)_{1,2} = 44.44$$

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{thn}$$

$$1.5 = 1.8 - |V_{GS3}| + 0.4$$

$$|V_{GS3}| = 0.7 \text{ V}$$

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV3})^2$$

$$100 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times \frac{W}{L} \times (0.2)^2$$

$$\left(\frac{W}{L}\right)_{3,4} = 50$$

$$(b) \quad A_v = g_{m1} (r_{o1} \parallel r_{o3})$$

$$g_{m1} = \frac{2I_{D1}}{V_{OV1}} = \frac{2 \times 100 \times 10^{-6}}{0.15} = 1.33 \text{ mS}$$

$$r_{o1} = \frac{1}{\lambda_n I_D} = \frac{1}{0.05 \times 100 \times 10^{-6}} = 200 \text{ K}\Omega$$

$$r_{o3} = \frac{1}{\lambda_p I_D} = \frac{1}{0.1 \times 100 \times 10^{-6}} = 100 \text{ K}\Omega$$

$$\therefore A_v = 1.33 \text{ mS} (200 \text{ K} \parallel 100 \text{ K})$$

$$= 1.33 \text{ mS} \times 66.67 \text{ K} = 88.67 \text{ V/V}$$

$$(c) \quad \text{Output voltage swing} = V_{DD} - |V_{OV4}| - V_{OV2} - V_{OV5}$$

$$= 1.8 - 0.2 - 0.15 - 0.25$$

$$= 1.2 \text{ V}$$

$$(d) \quad f_{3-dB} = \frac{1}{2\pi C_L R_{out}}$$

$$= \frac{1}{2\pi \times 2 \times 10^{-12} \times 66.67 \text{ K}} = 1.19 \text{ MHz}$$

$$(e) \quad f_{3-dB, closed} = f_{3-dB, open} (1 + \beta A)$$

$$= 1.19 (1 + 88.67) \text{ MHz} = 106.7 \text{ MHz}$$

$$f_{3-dB, closed} = \frac{1}{2\pi C_L \left(\frac{1}{g_{m2}} \parallel r_{o4}\right)} = 106.63 \text{ MHz}$$

$$(a) \quad I_{D6} = I_{D5} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV6})^2$$

$$120 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times 30 \times (V_{OV6})^2$$

$$\therefore V_{OV6} = 0.2 \text{ V}$$

$$V_{in,CM,min} = V_{OV5} + V_{GS1}$$

$$0.75 = 0.2 + V_{GS1}$$

$$\therefore V_{GS1} = 0.55$$

$$I_{D1} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{OV1})^2$$

$$60 \times 10^{-6} = \frac{1}{2} \times 200 \times 10^{-6} \times \frac{W}{L} (0.55 - 0.4)^2$$

$$\left(\frac{W}{L}\right)_{1,2} = 26.67$$

$$V_{in,CM,max} = V_{DD} - |V_{GS3}| + V_{thn}$$

$$1.45 = 1.8 - |V_{GS3}| + 0.4$$

$$|V_{GS3}| = 0.75 \text{ V}$$

$$I_{D3} = \frac{1}{2} \mu_p C_{ox} \frac{W}{L} (V_{OV3})^2$$

$$60 \times 10^{-6} = \frac{1}{2} \times 100 \times 10^{-6} \times \frac{W}{L} \times (0.25)^2$$

$$\left(\frac{W}{L}\right)_{3,4} = 19.2$$

$$(b) \quad A_v = g_{m1} (r_{o1} \parallel r_{o3})$$

$$g_{m1} = \frac{2I_{D1}}{V_{OV1}} = \frac{2 \times 60 \times 10^{-6}}{0.15} = 0.8 \text{ mS}$$

$$r_{o1} = \frac{1}{\lambda_n I_D} = \frac{1}{0.05 \times 60 \times 10^{-6}} = 333.33 \text{ K}\Omega$$

$$r_{o3} = \frac{1}{\lambda_p I_D} = \frac{1}{0.1 \times 60 \times 10^{-6}} = 166.67 \text{ K}\Omega$$

$$\therefore A_v = 0.8 \text{ mS} (333.33 \text{ K} \parallel 166.67 \text{ K})$$

$$= 0.8 \text{ mS} \times 111.11 \text{ K} = 88.89 \text{ V/V}$$

$$(c) \quad \text{Output voltage swing} = V_{DD} - |V_{OV4}| - V_{OV2} - V_{OV5}$$

$$= 1.8 - 0.25 - 0.15 - 0.2$$

$$= \underline{\underline{1.2 \text{ V}}}$$

$$(d) \quad f_{3-dB} = \frac{1}{2\pi C_L R_{out}}$$

$$= \frac{1}{2\pi \times 4 \times 10^{-12} \times 111.11 \text{ K}} = 358.10 \text{ KHz}$$

$$(e) \quad f_{3-dB, closed} = f_{3-dB, open} (1 + \beta A)$$

$$= 358.10 (1 + 88.89) \text{ KHz} = 32.19 \text{ MHz}$$

or

$$f_{3-dB, closed} = \frac{1}{2\pi C_L \left(\frac{1}{g_{m2}} \parallel r_{o4}\right)} = 31.95 \text{ MHz}$$